

Veda P. Yakkali

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EDUCATION

Rutgers University

Master of Science in Computer Science, Concentration in AI/ML

Relevant Coursework: Introduction to AI/ML, Brain Inspired Computing, Natural Language Processing

New Brunswick, NJ

Expected December 2026

GPA: **4.0**

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, PostgreSQL, C/C++, HTML/CSS, R, Bash

Frameworks: React, Spring Boot, Node.js, Next.js, Flask, AWS, Firebase, Supabase, FastAPI, Tailwind

Developer Tools: Git, Docker, Shell Scripting, Jira, GenAI Tools, Linux, VS Code, Eclipse, Vercel

Libraries: PyTorch, Pandas, NumPy, OpenCV, Matplotlib, TensorFlow, SciPy, NLTK, Jupyter, HuggingFace

Certifications: AWS Certified Cloud Practitioner, AWS Academy Graduate Machine Learning Foundations

EXPERIENCE

Research Assistant

Rutgers University

May 2024 – Present

New Brunswick, NJ

- Investigated YOLOv7 performance trade-offs on NVIDIA Jetson Nano, benchmarking energy-accuracy metrics to optimize real-time wildfire detection on edge devices
- Optimized system memory by **18%** and inference throughput by **14%** by implementing spatial masking and temporal delta calculations to eliminate redundant computation
- Accelerated end-to-end video to inference process by **32%** by offloading color conversion and YOLO preprocessing to the GPU using **NVIDIA NPP** and **PyTorch**
- Architected a modular deployment pipeline by containerizing preprocessing and inference modules separately in **Docker** for reproducible execution on edge hardware

Full Stack Developer

Rutgers Enactus

September 2022 – May 2025

New Brunswick, NJ

- Collaborated on a full-stack platform using **React** and **Firebase** serving **100+** users by enabling job discovery and resource management workflows
- Developed a React-based login interface integrated with Firebase Authentication, enabling secure user access and account-linked resume storage
- Built an AI-driven career assistant via **OpenAI API** integration, automating **70%** of initial user inquiries through context-aware retrieval of platform resources
- Optimized backend data access by restructuring Firebase query patterns, reducing database read costs by **35%**

PROJECTS

Bitcoin Fraud Detection | Java, Python, PyTorch, scikit-learn, Pandas, NumPy, Matplotlib

December 2025

- Built a fraud detection system for a Bitcoin transaction graph consisting of **200K+** nodes and **234K+** edges, achieving **95.8% accuracy** through Random Forest and Logistic Regression models
- Developed a Graph Contrastive Learning (GraphCL) pipeline to perform semi-supervised learning on 166-dimensional transactional features, reaching a **88% recall** rate for illicit activity detection

Point of Care Automatic Arrhythmia Detection | Python, PyTorch, SciPy, Pandas

October 2025

- Developed a Spiking Neural Network (SNN) model for heart arrhythmia detection, achieving **94.6%** diagnostic accuracy on the MIT-BIH clinical dataset
- Reduced computing cost by **82%** relative to a CNN baseline by implementing Leaky Integrate-and-Fire (LIF) spiking neurons for resource constrained devices

Formula One Fantasy | Java, Python, PostgreSQL, React, Spring Boot

August 2025

- Built a full-stack analytics platform using **React** and **Spring Boot** to deliver race and driver statistics through a suite of **RESTful APIs**
- Engineered a custom data pipeline using **Pandas** to clean and organize F1 datasets into a **PostgreSQL** database for efficient backend querying